The 1997 draft admits more than the certified full-prime-order set

Date: 2026-10-24 UTC. **Question:** How many seed-derived curves in the retained *positive exact-order* records satisfy the November 1997 X9.62 draft's near-primality procedure with its example bound `l_max=255`? **Verified result: 99 distinct seed positions**: the prior 69 with prime full order and **30 additional positions** with a 255-smooth cofactor and a proven large prime subgroup. All 99 have independent subgroup-point/Hasse certificates. This is a **lower bound** on the reconstructed prefixes' draft-compatible set, because 10,083 other seeded positions have censored exact-order outputs on eligible coefficient roots. It is not a count of attempts by the original program.

Dated rule, exact set and controls

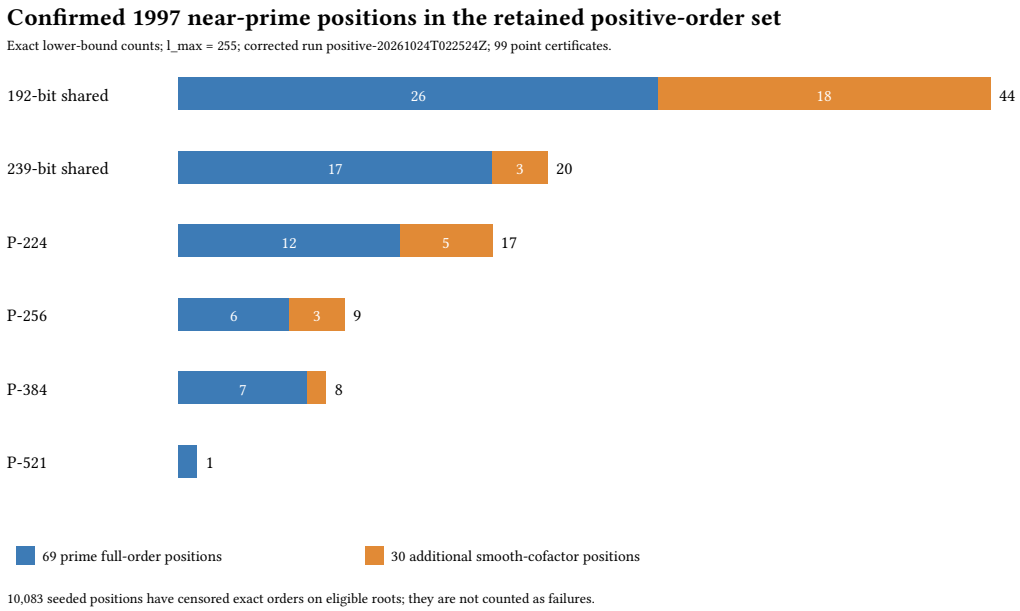
Retrieved: 17 November 1997 ANSI X9.62 working draft, PDF pp. 61–65, SHA-256 `0c231d5176fcf0b239b44732213d281b518a0e16cc6f13f7bd4039ec99807b6e`. Section F.2.2 divides the group order by all prime powers through an operator-supplied `l_max`, then accepts a sufficiently large probable-prime residual. Section F.3.2 gives `255` as a typical trial bound and requires `r_min > 2^160`; it then checks MOV and anomalous conditions and finds a point of prime order. The text does not mandate cofactor one or explain the historical seed schedule. This round fixes **`l_max=255` and `r_min=2^160+1`** as a reproducible reference setting, not as a recovered private setting. The same 99 also satisfy the later `n>4√p` lower bound.

Internal exact inputs: the six `orders_<timestamp>.txt` files and validated census cross-check, whose digests are bound in the corrected run receipt. The census has 11,042 positions with a seed-derived coefficient pair in the six prefixes. The corrected interpretation gives 959 positions with at least one positive exact order on a seeded root; 10,083 have no such retained order. Among the positive records, 1,374 root branches were tested. A zero from the original `ellsea(E,1)` early-abort mode is **censored**; it is not an exact composite order or proof that the draft would reject that position.

Measured and independently checked: run.py removed all factors of primes at most 255 from each positive exact order, checked the residual with PARI/GP `isprime`, and applied the subgroup lower bound, MOV-20 and anomalous exclusion. It retained exact per-branch rows, GP input and raw output. verify.py separately re-read every archived order, repeated the divisions, bounds and modular checks, and cross-checked every GP classification with SymPy 1.14.0. SymPy uses probable-prime methods at these sizes; GP `isprime` is the primary primality result.

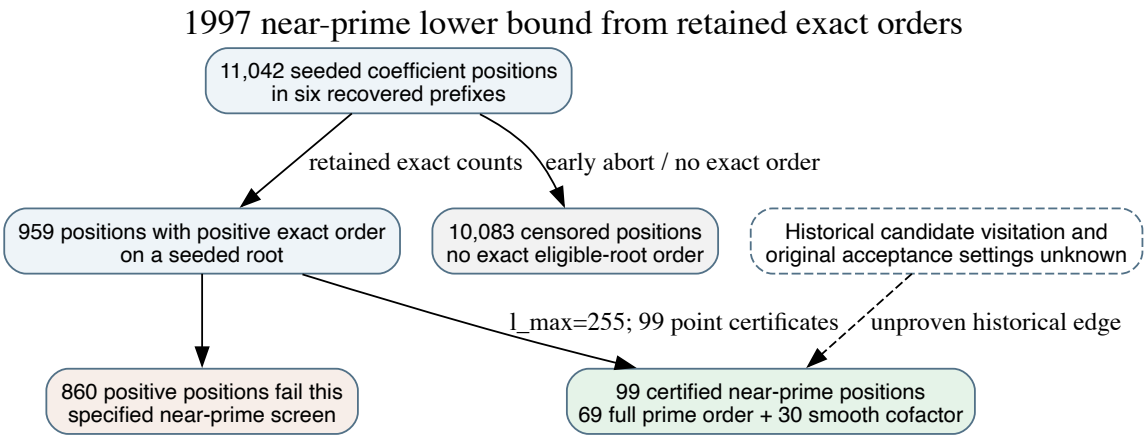
The exact 99 seed positions were then checked under the frozen point-certificate protocol. The successful Sage run chose a deterministic x from 1 through 32 for every position and produced a nonzero point `P` with `[n]P=0`. Every residual `n` passed Sage's proof-enabled primality check and `n²>16p`; Hasse's interval therefore contains only one possible multiple of `n`, the archived `u=h*n`. The 99 full parameter and point records identify exact field, seed, coefficient, subgroup order, cofactor and chosen point. A separate pure-Python affine replay checked all 99 curve equations and `[n]P=0`, including all ten published and eleven named omission controls; its verification record binds the raw file.

Reconstructed stream	Certified full prime order	Additional 255-smooth cofactor	Confirmed near-prime total	Censored positions
192-bit shared	26	18	44	2,779
239-bit shared	17	3	20	1,430
P-224	12	5	17	3,289
P-256	6	3	9	815
P-384	7	1	8	1,246
P-521	1	0	1	524
Total	**69**	**30**	**99**	**10,083**



Blue denotes full prime order; orange adds the 30 smooth-cofactor positions. Exact counts within the retained positive-order set have no sampling error. Editable chart source.

The 30 new positions are unpublished, so the reference-setting lower bound is **10 published and 89 unpublished**. Seven of the 30 qualify on the census's selected root, and 23 only on its other seed-derived root in fields with $p \equiv 3 \pmod 4$. Their minimum smooth cofactors range from 9 to 1,095,881; every prime factor of each is at most 255. These orders would fail a cofactor-one preference, which all ten published prime-field curves satisfy, but the dated X9.62 near-prime procedure does not impose that preference. All 99 exceed $4\sqrt{p}$, so adding that later minimum changes no count in this observed set.



Solid edges show source-bound calculations. The dashed edge requires historical code or logs. Editable diagram source.

Corrections and inference boundary

The first near-prime run yielded 130 positions but incorrectly counted P-224's `nt` column as the other seed-derived coefficient sign. It is actually the order of a different quadratic twist when `p ≡ 1 (mod 4)`; the two seed-derived signs are isomorphic and share the first-column order. The run, its verifier and exact first launcher are retained; AMENDMENT.md records why its 130-position figure is invalid. The corrected run has 99 positions. A first point-certificate run stopped after 17 positions because its launcher paired a census order with the wrong element of a **numerically sorted** coefficient-root list. A direct two-root Sage diagnostic, reproducible with diagnose_root.py, identified the inversion; POINT_AMENDMENT.md freezes the corrected root binding and preserves the failed attempt. These were input-mapping errors, not evidence against a curve.

Derived: Under the stated reference setting and retained exact orders, the known eleven omissions are a still narrower subset of 89 unpublished near-prime possibilities. This arithmetic establishes neither historical visitation nor a shared hidden weakness. A historical explanation must first show which root and cofactor policies the original program used and why it passed over the eleven prime-order candidates remaining after those policies. The original program or logs remain the decisive acquisition target; a mathematical property should be declared before testing and predict all relevant controls. The open selection-rule problem remains unresolved. This result adds no verified ECDLP transfer or cost edge; the existing canonical graphs were checked and remain unchanged.

Reproduction: Run run.py into a fresh directory under the frozen protocol plus amendment; check its exact output with verify.py. Reproduce the 99 point records using `gtimeout 120 sage -python research/nist_seed_1997_nearprime_20261024/verify_points.py` under POINT_PROTOCOL.md plus POINT_AMENDMENT.md, then run verify_point_certificates.py on the resulting directory. The checked run and version details, source hashes, controls, measured local time and machine resource figures are in their receipts. Source provenance: draft `retrieved`; censored/positive point-count records `internal`; trial divisions, 99 point certificates and independent replays `checked`; historical selection `unknown`.